



CENTRE FOR ADVANCED STUDIES

Dr. A.P.J Abdul Kalam Technical University, Lucknow

NEWSLETTER - JANUARY 2022

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डा० ए०पी०जे० अब्दुल कलाम प्राविधिक विश्वविद्यालय, उत्तर प्रदेश, लखनऊ

Dr. A.P.J. Abdul Kalam Technical University, Uttar Pradesh, Lucknow

(Formerly Uttar Pradesh Technical University)

MESSAGE FROM THE VICE CHANCELLOR



Prof. Pradeep Kumar Mishra

Vice Chancellor

Dr. A.P.J. Abdul Kalam Technical University

I am glad to learn that Centre for Advanced Studies, Lucknow is publishing the first edition of the Newsletter in January 2022. I am very happy to learn that the faculty and students of Centre for Advanced Studies are involved in good quality academic and research work and other important activities. I wish the centre works in areas that are relevant to current social and national importance. I also wish Centre for Advanced Studies will work towards creation of good research ecosystem for doing research and innovation activities and also provide a platform to other Institutions to use these facilities in collaboration. With the start of the Incubation Hub in the Centre for Advanced Studies an acceleration is expected in the areas of innovation and start-ups that may provide a good platform to the young innovators. I hope that the newsletter will become a widely popular chronicle and my best wishes for this edition.



CENTRE FOR ADVANCED STUDIES

Dr. A.P.J Abdul Kalam Technical University, Lucknow

Sector- 11, Jankipuram Extension, Lucknow, UP 226031



MESSAGE FROM THE DIRECTOR



Prof. M.K. Dutta

Director, **Centre for Advanced Studies**
Dr. A.P.J. Abdul Kalam Technical University

It gives me immense pleasure to present the inaugural issue of the Newsletter of Centre for Advanced Studies. This newsletter includes some of the events and activities of the centre. Centre for Advanced Studies is an in-campus research driven institute established by Dr. A.P.J Abdul Kalam Technical University Lucknow to impart state of the art education to post graduate students and PhD Scholars and to facilitate quality research work in the emerging areas of Engineering and Technology. The institute offers M.Tech and PhD programs in the disciplines of Computer Science and Engineering, Mechatronics, Nanotechnology, Manufacturing Technology and Automation, and Energy Science and Technology. The centre aspires to provide a stimulating platform to researchers and academicians for creating and disseminating research based knowledge and technologies for the development of State and Country. The high quality labs and infrastructure facilitates high quality research and brings in lots of hope for futuristic work. I wish all the very best to the faculty, staff and students and hope they will continue to work with the same enthusiasm and put in more efforts in achieving the objectives of the Centre. My sincere thanks to the Editorial team of the Newsletter and wish the newsletter gets wide readability.

ABOUT THE INSTITUTE

Centre for Advanced Studies is an in-campus research driven institute established by Dr. A.PJ Abdul Kalam Technical University Lucknow to impart state of the art education to post graduate students and to facilitate quality research work in the emerging areas of Engineering and Technology. The institute offers M.Tech. and Ph.D programs in the disciplines of Computer Science and Engineering, Mechatronics, Nanotechnology, Manufacturing Technology and Automation, and Energy Science and Technology. Established by the Uttar Pradesh State Government in 2017 with an objective to provide a stimulating platform to research scholars and academicians for creating and disseminating research based knowledge and technologies for the development of State/Country. The Institute is climbing up consistently on the path to visibility across the globe. In a short span of time, significant progress has been made with quality education, impactful research, publications and patents, funded projects, training and placement. The University is also making constant efforts to create a healthy environment for meaningful research outcomes, to mentor affiliated Institutions with an establishment of world class laboratories and facilities in the Institute, and to enhance the knowledge of faculty and students with the latest technologies and developments through training and education.

Centre for Advanced Studies networks with the existing institutions, laboratories, universities and colleges in the neighbourhood in order to make optimal utilization of resources, both in terms of expensive laboratory facilities as well as teaching talents. It is an unique institution in the country where engineering & technology education is totally integrated with state-of- the-art research, therefore, an Integrated postgraduate degree programme in engineering is offered following graduation. The Institute is currently offering M.Tech and PhD in 5 programs listed below along with their focus areas.



Computer Science & Engineering

- Artificial Intelligence(Machine Learning and Deep Learning)
- Cyber Security
- ICT (Information & Communication Technology)
- Data Science & Analytics
- Internet of Things (IoT)
- Computer Vision
- Cloud Computing



Mechatronics

- Robotics
- PLC & SCADA
- AI based systems
- Drones and Aerial systems

Manufacturing Technology & Automation

- Industry 4.0
- 3D Printing
- Micro and Nano
- Manufacturing
- Automation



Nanotechnology

- Nanomaterial synthesis and characterization
- Nano-Fabrication
- MEMS/NEMS Devices
- Microfluidics

Energy Science and Technology

- Unconventional/Renewable Energy
- Energy conversion and Storage System
- Alternate Energy
- Energy Management



Artificial Intelligence based Intelligent Stethoscope: A low-cost device based on body auscultation for early medical diagnosis of heart, lung and prenatal health

Cardiovascular and pulmonary diseases are critical diseases and need to be diagnosed as early as possible. There is a lack of medical professionals in remote areas to diagnose these diseases. An Artificial Intelligence based automatic intelligent stethoscope is developed to diagnose cardiac, pulmonary and prenatal disorders to minimize human intervention. The proposed system involves a convolutional neural network (1D-CNN) model to diagnose the cardiac disorders from the PCG signal and lungs disorder from pulmonary signals because of its high accuracy and robustness. To improve the accuracy in a noisy environment and make the method robust, the proposed method has used data augmentation techniques for training and multi-classification of cardiac diseases. The proposed model is ported into small computing devices to make a stand-alone diagnostic tool that may be of help in remote primary health care centers. The developed prototype contains a stethoscope, sound card, display screen, DSP processor, and power bank. A user friendly

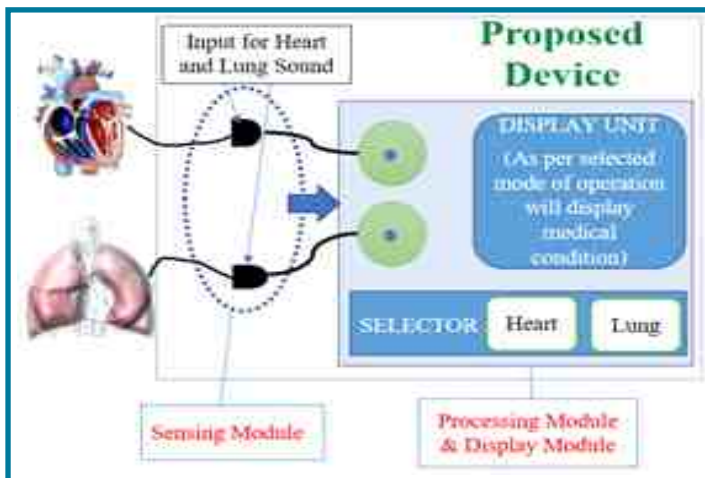


Fig. 1 Flowchart of the proposed method.

GUI application is developed which is based on python for viewing and analysis of generated reports. The proposed method is non-invasive, efficient, robust, and has low time complexity making it suitable for real-time applications.

This device uses the technology based on artificial intelligence where machine learning and deep learning is used for training the model for heart, lungs and prenatal sound. The developed device integrates a low-cost DSP processor with a screen

and stethoscope. Its aim is to diagnose heart, lungs and prenatal health and classify the patient's (heart, lungs and prenatal sound) as normal or abnormal with type of disease.

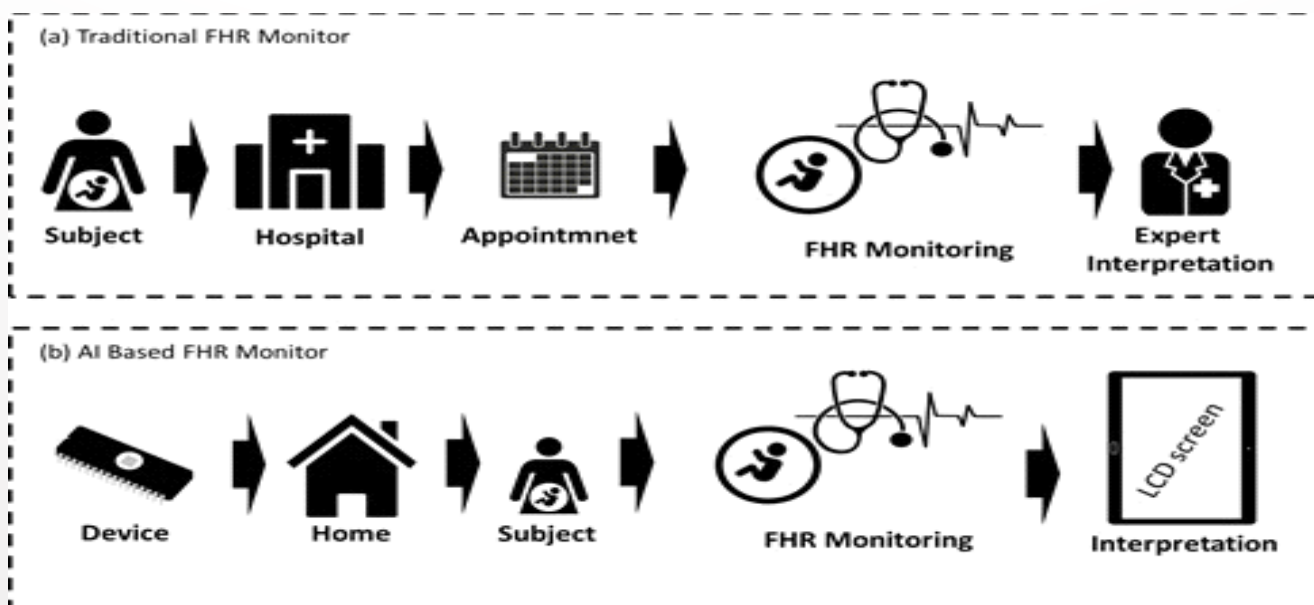


Fig. 2: Fetal Heart rate monitoring during pregnancy

Fetal heart rate (FHR) is used to monitor the fetal state by obstetricians as a screening tool. Common guidelines for visual interpretation of FHR signals results in significant subjective variability due to the fetal physiological dynamics complexity. Automated diagnostic technology can assist obstetricians in medical decisions based on artificial intelligence and also can be an automatic diagnostic tool for primary health care centres and remote areas.

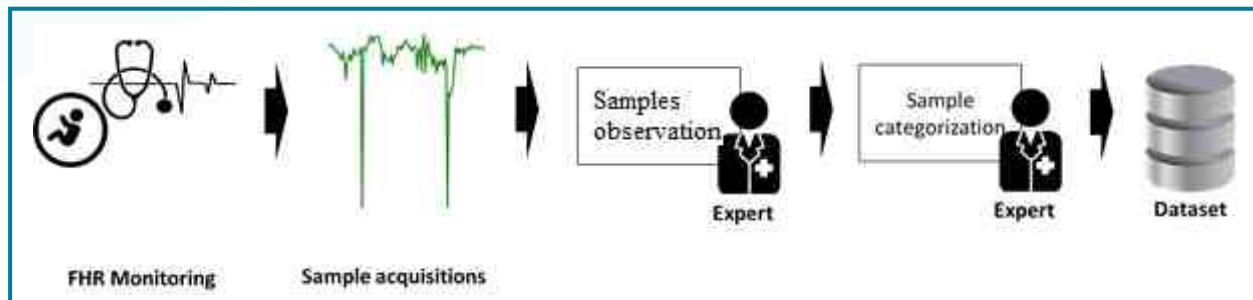


Fig. 3: CTU-UHB dataset preparation

It is a low-cost, real time and small portable intelligent device. These type of devices are not available in the market. This device is absolutely new where automatic diagnosis of heart, lungs, prenatal health can be performed without trained manpower. This device will have a great impact in the society due to its small size and low budget. It is

based on artificial intelligence where machine learning and deep learning is used for training the model for heart, lungs and prenatal sound. It can be used for the screening purpose in the remote locations, primary health care centers, health care camps, medical clinics and personal use.

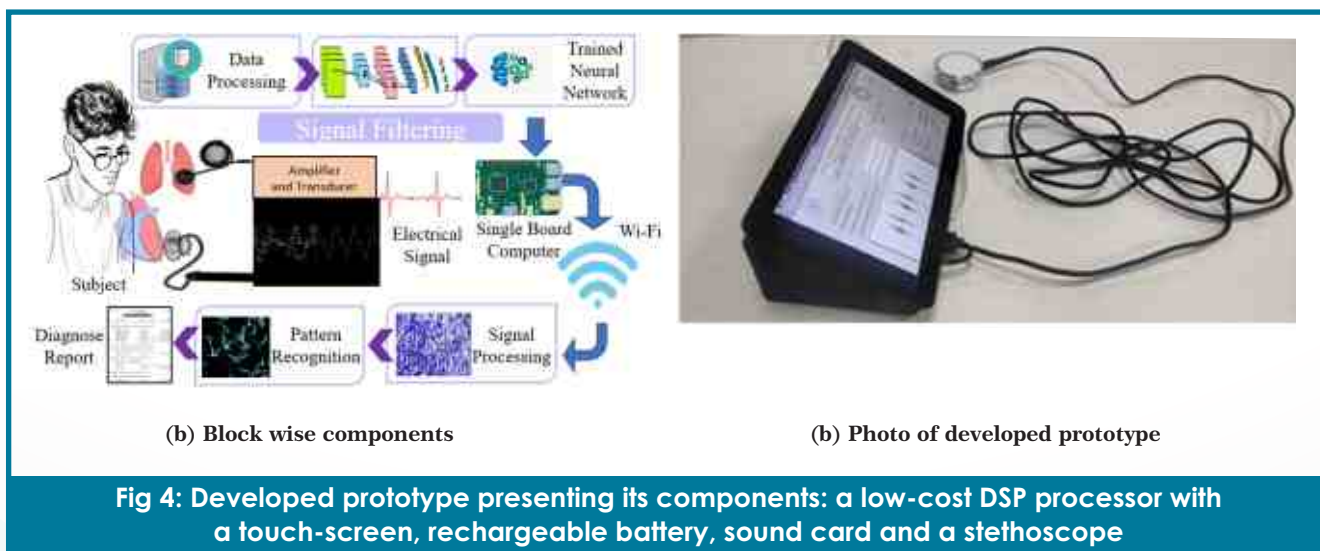


Fig 4: Developed prototype presenting its components: a low-cost DSP processor with a touch-screen, rechargeable battery, sound card and a stethoscope

Specifications:

- Device dim approximately 19cm*11cm*3cm
- Ultra low power ~ 1W
- 16bit per sample,
- 8000 Hz sampling rate
- 128kbps Bitrate
- Gaussian butter worth filter
- Proper GUI user interface
- 10 Finger Capacitive Touch
- Diagnose heart, lungs and health

Working Process:

- Fill patients details (Name, Age, Gender)
- Place stethoscope.
- Press record button on GUI
- Record and processing time: 5-10 sec
- View graph
- Press generate report
- View detailed report

The device also has a Graphical User Interface (GUI) facility on which details of the patient can be entered. It generates a report stating the patient's heart condition. Prototype is practically usable and affordable, making it well suited for medical screening at primary health care centers and remote areas in India.

"It is a low-cost electronic hardware integrated with the proposed methodology which can perform in real-time and can achieve high accuracy and reliability. It can be used to support the expert decision and as an automatic stand-alone diagnostic tool which can be a potential breakthrough within the Biomedical Engineering in India." – Prof. M.K. Dutta (PI)

Research Team:



Neeraj Baghel
Junior Research Fellow



Juwairiya Siraj Khan
Junior Research Fellow



Anjali Yadav
Junior Research Fellow

Patents from the project:

- (A) "A System and Method for Machine Learning Based Automatic Detection of Abnormality from Cardiac Sound" Application number: 201911033584 Dated: 21.08.2019, Status: Applied and under Examination
- (B) "Artificial Intelligence based System and Method for Multi-Class Classification of Heart Diseases", Allotted Application number 201911047767 Dated: 22.11.2019, Status: Applied and under Examination
- (C) "Automatic Detection of Lungs Disease using Machine Learning from Respiratory Sound" Application number: 202011009043 Dated: 03.03.2020, Status: Applied and under Examination

Publications from the project:

1. Anjali Yadav, Anushikha Singh, Malay Kishore Dutta & Carlos M. Travieso "Machine Learning Based Classification of Cardiac Diseases from PCG recorded Heart Sounds" Neural Computing and Applications, Springer Verlag Publishers, DOI - 10.1007/s00521-019-04547-5, December 2020, SCI indexed Impact Factor – 5.606.
2. Neeraj Baghel, Vivek Nangia and M.K.Dutta "ALSD-Net: Automatic Lung sounds Diagnosis Network from Pulmonary Signals" Neural Computing and Applications, DOI: 10.1007/s00521-021-06302-1, 2021, Springer Nature Publishers, SCI indexed impact factor 5.606.
3. Neeraj Baghel, Radim Burget, Malay Kishore Dutta "1D-FHRNet: Automatic Diagnosis of Fetal Acidosis from Fetal Heart Rate Signals", Detection in Children using Deep Neural Network" Biomedical Signal Processing and Control, Elsevier Publisher, doi.org/10.1016/j.bspc.2021.102794, SCI indexed Impact Factor – 3.88.
4. Neeraj Baghel, Malay Kishore Dutta & Radim Burget, "Automatic Diagnosis of Multiple Cardiac Diseases from PCG Signals using Convolutional Neural Network" Computer Methods and Programs in Biomedicine, Volume 197, doi.org/10.1016/j.cmpb.2020.105750, Elsevier Publishers, 2020, SCI indexed, Impact Factor – 5.428.
5. Juwairiya Siraj Khan, Manoj Kaushik, Anushka Chaurasia, Malay Kishore Dutta, Radim Burget, "Cardi-Net: A Deep Neural Network for classification of Cardiac disease using Phonocardiogram Signal" – Computers Programs and BioMedicine, Elsevier Publications – Submitted and under Review



"It is a low-cost electronic hardware integrated with the proposed methodology which can perform in real-time and can achieve high accuracy and reliability. It can be used to support the expert decision and as an automatic stand-alone diagnostic tool which can be a potential breakthrough within the Biomedical Engineering in India"

Prof. M.K. Dutta Principal Investigator

An AI-Enabled Assistive Device for Visually Impaired People

The major sensory organ of a person is their eyes. One glimpse around us is enough to make us realize the importance of visual information. Most of this information is inaccessible for the blind and visually impaired people which inhibit their independence. Visually impaired people face numerous difficulties in their daily life, and technological interventions may assist them to meet these challenges. Science and engineering can make technical interventions to the lives of visually impaired people in making them independent to navigate and perceive objects around them. Many devices have been proposed to assist a visually challenged person, but most of the devices focus either on object detection through computer vision or on obstacle detection through different sensors, such as GPS, distance sensors, etc. However, the effective utilization of sensor-based technologies and computer vision could result in a highly efficient and supportive device, to make them aware of the surroundings. Hence, a prototype of

artificial intelligent fully automated assistive device is designed for visually impaired people to perceive the objects in the surrounding and provide obstacle-aware navigation, where auditory inputs are given to users in real-time and gives better understanding to the visually impaired person about their surroundings. Images of objects that are highly relevant in the lives of the visually challenged are trained using deep learning neural networks. Augmentation and manual annotation are performed on the dataset to make the system robust and free from overfitting. Both sensors and computer-vision based techniques are integrated to provide convenient a travel-aid to visually impaired people, through which a person can perceive multiple objects, detect obstacles and avoid collisions. The auditory information that is conveyed to the user after scene segmentation and obstacle identification is optimized to obtain more information in less time for faster processing of video frames.



Block Diagram of Artificial Intelligence based Assistive Framework for Visually Impaired People

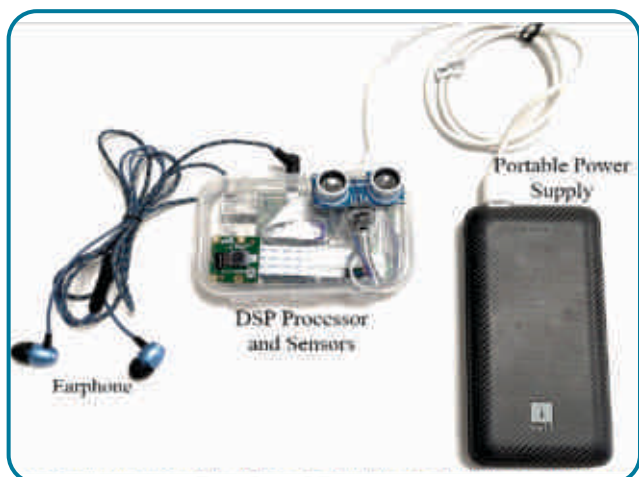
The proposed module consists of a DSP processor with a distance sensor, camera, and power supply. Speakers or headphones are associated with the DSP processor to perceive predictions as an audio prompt. Once the CNN was trained with the dataset, the final trained model was equipped in the object detection framework. A live video feed was associated to the framework and image frames were subsequently captured. Captured frames were

pre-processed and fed into the trained model, and if any object which was trained with the model was detected, a bounding box was drawn around that object and a respective label was generated for that object. Once all objects were detected, the text label was converted into speech, or a respective audio label recording was played, and subsequently, the next frame was processed.

All previous inventions and research works for blind or visually impaired people which use ultrasonic sensors to detect obstacles define their range and play a warning sound whenever an obstacle comes across the sensor. When the calculated distance through the ultrasonic sensor is below the threshold value, the device makes an acoustic warning or vibrates, but it can be irritating for a visually impaired person who is standing in a crowd and repeatedly listening same prompt or continuous vibrations. So, one of the objectives of the proposed system is to differentiate between trained objects and obstacles.

The system first analyses the current frame for object detection—if an object is detected which means the object is in front of the device, then there is no need for searching another obstacle. If

no objects are identified in present frame, then it takes input from an ultrasonic sensor regarding the distance from the object, and if the calculated distance is less than the threshold,



Developed Prototype

Working Process:

- Switch on the device.
- Swipe hand in front of the device to select different modes.
- Device capture image from video streaming
- Perform Object detection and classification using trained model
- Device give audio output of object name or obstacle.
- Device will read the text on selection of TEXT READER mode
- On not selecting any mode device will get switched off.

Different modes are designed in the device to provide wider assistance such as indoor, outdoor or text-reader mode. The indoor mode has a smaller threshold value for the distance of obstacle compared to the outdoor mode. Outdoor mode also has an image enlarge function, so that far objects can be detected early. For example, a car at far distance can be easily detected when enlarged, because after enlarging the image, the number of pixels is increased, and it becomes an easy task to detect that car. This feature provides an audio prompt when the object is at distance, and helps user to be alert to their surroundings, as early detection is crucial in case the user is outside and especially in those scenarios such as when a car is coming towards the person. The text reader mode can be used efficiently where the user has a necessity to read, such as when reading a book, a restaurant menu, etc. To read text, Optical Character Recognizer (OCR) is used after preprocessing the input image frame. Face recognizer can also be associated with the device, where users can identify known persons and family members, which will help in them to be social and secure.



New modified blind assistive prototype

The whole framework is standalone and designed for low-cost processors, so that the visually impaired can use it properly without internet connectivity. The detected output is also optimized to ensure faster frame processing and a greater extraction of information—i.e., count of objects—in a shorter time period. This prototype can easily differentiate between obstacles and known objects. The innovation can make a significant contribution to assist visually impaired people compared to previously developed methods, which were only focused on obstacle detection and location tracking with the help of basic sensors without use of deep learning. With the use of this device prototype, users can interpret more about the things going around and receive obstacle-aware navigation as well.

“Disability is only a state of mind. With the use of adaptive equipment and training with it, the mobility can be made easier. Hence, sensors and computer-vision based techniques are integrated to provide convenient a travel-aid to visually impaired people, through which a person can perceive multiple objects, detect obstacles and avoid collisions.”

- Prof. M.K. Dutta (Principal Investigator)



Research Team:



Rakesh Chandra Joshi
Research Scholar



Saumya Yadav
Junior Research Fellow



Abha Upadhyay
Junior Research Fellow



Anushka Chaurasia
Junior Research Fellow

Patent from the project:

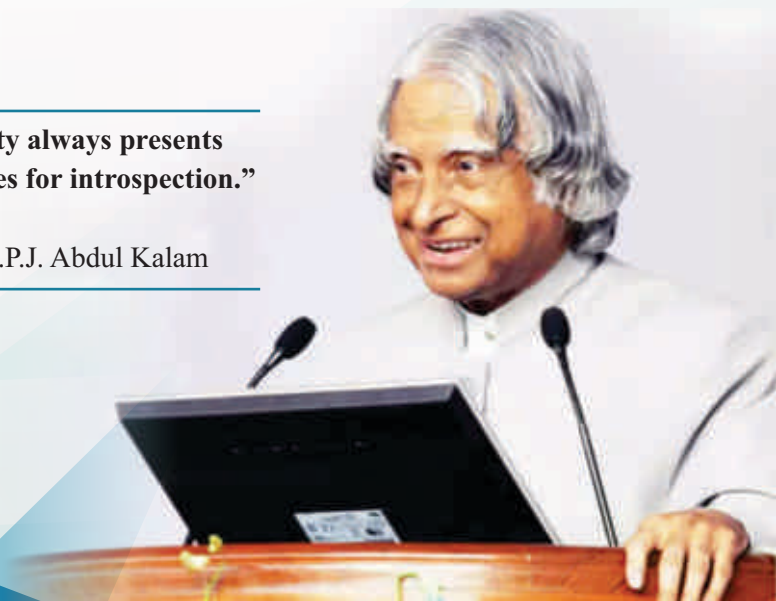
“Multi Mode Real-Time Object Detector and Obstacle Aware Assistive System for Visually Impaired Person (VIP)”, Application No.- 201911047766, Dated - 22.11.2019, Status – Applied and Under Examination

Publications from the project:

- 1) Joshi, R.C.; Yadav, S.; Dutta, M.K.; Travieso-Gonzalez, C.M. Efficient Multi-Object Detection and Smart Navigation Using Artificial Intelligence for Visually Impaired People. Entropy 2020, 22, 941.
- 2) Yadav, R. C. Joshi, M. K. Dutta, Martin Kiac, Pavel Sikora, “Fusion of Object Recognition and Obstacle Detection approach for Assisting Visually Challenged Person” 2020 43rd International Conference on Telecommunications and Signal Processing (TSP), July 7-9, 2020, Milan, Italy.
- 3) R. C. Joshi, S. Yadav and M. K. Dutta, "YOLO-v3 Based Currency Detection and Recognition System for Visually Impaired Persons," 2020 International Conference on Contemporary Computing and Applications (IC3A), Lucknow, India, 2020.
- 4) D. Dahiya, H. Gupta and M. K. Dutta, "A Deep Learning based Real Time Assistive Framework for Visually Impaired," 2020 International Conference on Contemporary Computing and Applications (IC3A), Lucknow, India, 2020.
- 5) Lalita Jaiswar, Anjali Yadav, Malay Kishore Dutta, Carlos Travieso-González, Luis Esteban-Hernández, “Transfer Learning based Computer Vision Technology for Assisting Visually Impaired for detection of Common Places” 3rd International Conference on Applications of Intelligent Systems 2020 (APPIS 2020) Las Palmas de Gran Canaria, Spain, ACM Digital Library. DOI: 10.1145/3378184.3378215
- 6) Saumya Yadav, R. C. Joshi, M. K. Dutta, R. Burget. “Artificial Intelligence based Robust and Real-time Assistance for Visually Impaired People”- Assistive Technology (The Official Journal of RESNA), Taylor & Francis Publishers- Submitted and under review.

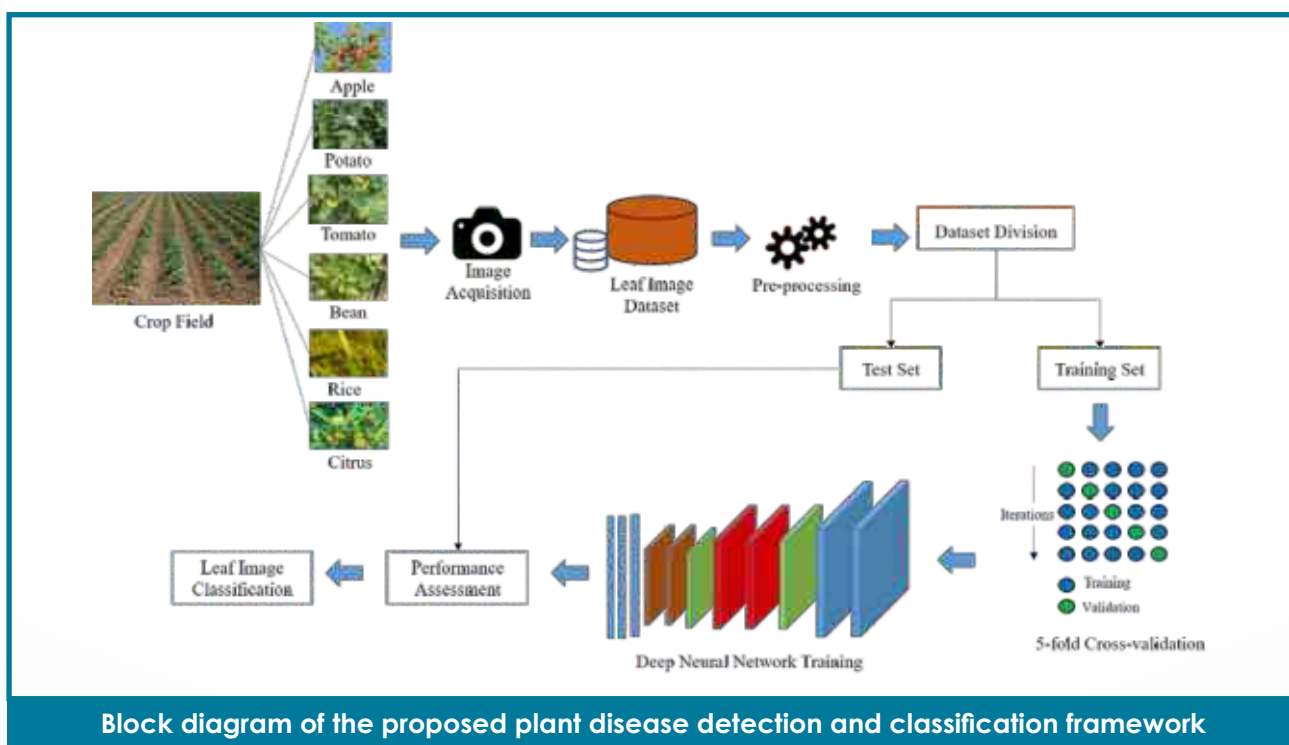
“Adversity always presents opportunities for introspection.”

— Dr. A.P.J. Abdul Kalam

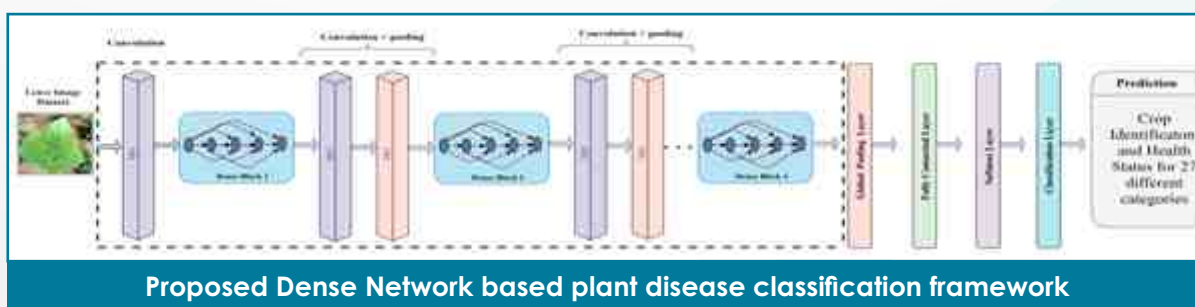


Artificial Intelligence in Agriculture: Efficient and Automated Plant Disease Detection and Classification

Students of Centre for Advanced Studies have jointly developed a deep-learning and Machine learning-based approach for plant disease detection and classification of leaf images. Plant diseases are one of the significant causes of the loss of crops. They can affect the quality of crops and dwindle their production. It can lead to famines and a rise in unemployment in the agriculture sector. Plant diseases are one of the major concerns in the agricultural domain and their automatic identification is very crucial in monitoring the plants. Dense Convolutional neural network architecture is trained on an image dataset (laboratory and on-field conditions) of six crops with multiple resolutions in 27 different categories from multiple countries. Images have several inter-class and intra-class variations with complex and challenging conditions that have been addressed in this dense neural network. Five-fold cross-validation and testing on unseen data are done for an exhaustive evaluation of the trained model in various parameters. The experimental findings demonstrate that an average cross-validation accuracy of 99.58% and average test accuracy of 99.199% is obtained on unseen images with complex background conditions. The processing time to process a single plant leaf image is 0.016 seconds with significant accuracy which signifies its real-time performance.



All layers are directly connected in a feedforward manner with each other layer in each dense block. All the feature maps at each layer gathered from preceding layers are separately considered and given as input after concatenating in a form of single tensor.



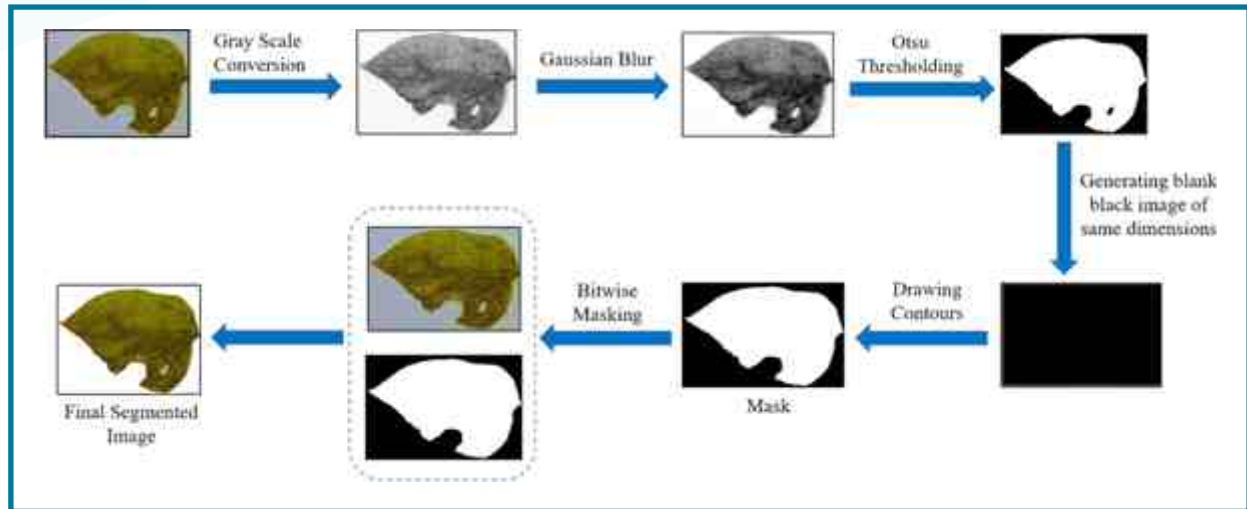
In another work in this domain, a novel deep learning model VirLeafNet is developed by research team to identify and diagnose the virus infected leguminous plant, *Vigna mungo* which is grown largely in the Indian subcontinent. *Vigna mungo* also referred to as

Urad Bean or Black Gram is a kind of bean. The early detection of the disease with this framework can help the farmers and necessary actions can be taken on time. Otherwise, the viral infections could lead to low yield of the crop resulting in high economical loss.

Due to viral infection, some properties of the leaf image changes but the pattern is very random throughout the leaf structure. Hence, it is quite challenging to make an automatic disease detection method and perform the detection tasks in real-time.

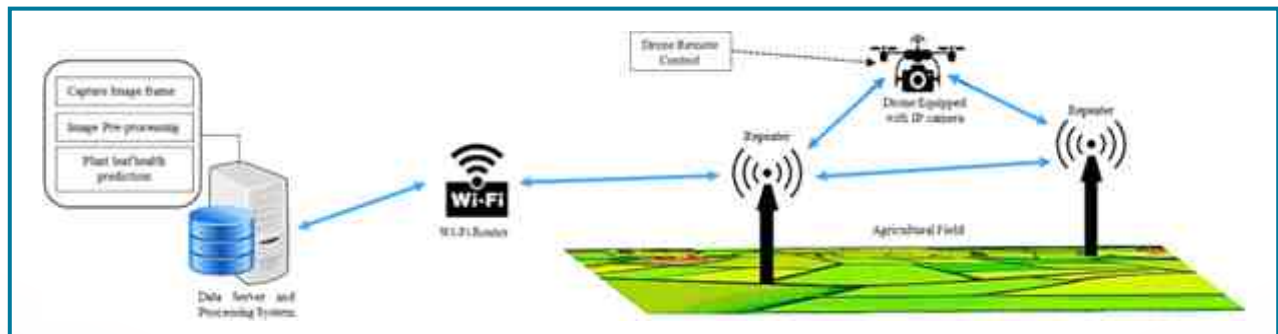
time. Otherwise, the viral infections could lead to low

yield of the crop resulting in high economical loss. Due to viral infection, some properties of the leaf image changes but the pattern is very random throughout the leaf structure. Hence, it is quite challenging to make an automatic disease detection method and perform the detection tasks in real-time.



Process of Leaf Segmentation

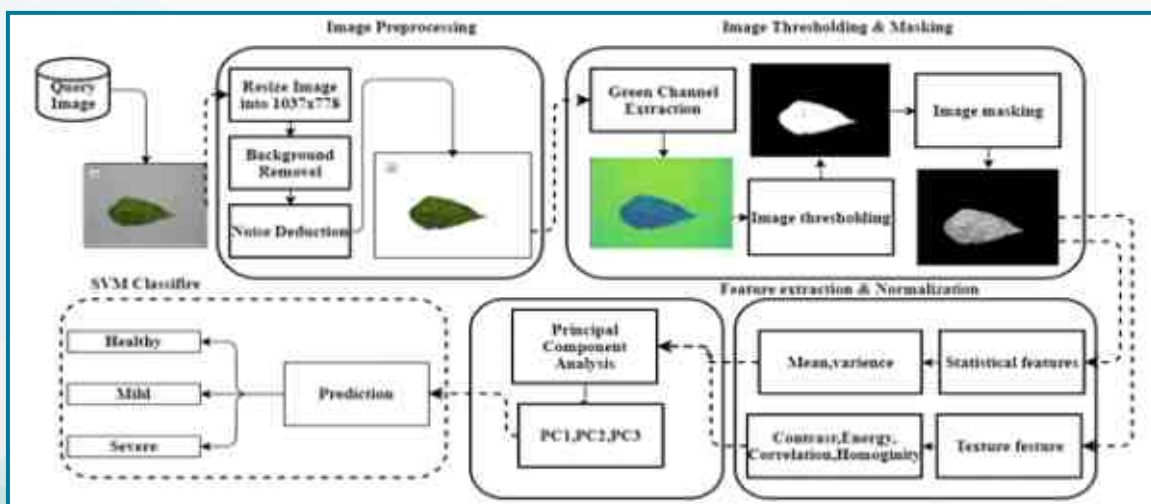
The proposed methodology can be integrated with drones for wider crop area analysis. The proposed method is completely automatic, non-destructive and quickly classifies the leaf images of different categories in real-time.



Drone integration to analyses the large crop area

Another machine learning based innovation for Automatic Diagnosis of Chlorosis in Vigna Mungo Plant" is done for early detection of Chlorosis problem in Vigna Mungo plant. The detection is based on a leaf pattern that can use to predict different problem related to the Vigna mungo plant which is necessary

for proper growth of the plant. The method uses Machine learning for identifying the severity of chlorosis in Vigna mungo plant. The method can be integrated with drones to surveillance the large cultivated fields in real-time. This type of system can be highly beneficial for farmers with large crop fields.



Machine learning method for prediction of plant leaf health

Developed Products for Prevention and Protection from COVID-19

Dr. Anuj Kumar Sharma and his team developed many useful products for prevention and protection from COVID-19 virus during first wave of COVID-19 pandemic in India. He filed patent for few of the developed products.



The sanitizing tunnel is developed for sanitizing the personal staff etc. at the entrance of any building or premises. The developed sanitizing tunnel is installed at Rajbhavan Uttar Pradesh.



The sanitizing tunnel is for sanitizing the luggage. This tunnel can be kept at the entry gate of railway stations, airports, post office or at any premises where luggage handling is major concern.



The sanitizing chamber is for sanitizing the personal protection equipment (PPE), which is used by doctors and paramedical staff in hospitals or even outside. There is shortage of PPE kits and even the PPE kits used by the staff in normal wards can be reused after getting it sanitized with appropriate technology.



This is simple to operate and economical solution to dispense the sanitizer while applying a little effort, without touching the nozzle of sanitizer bottle.



The universal portable sanitizing is for sanitizing the small items including electronic items like wrist watch, spectacles, ring, stethoscope, wallet etc. carried by doctors and paramedical staff in hospitals or even outside



This machine is equipped with a sensor which dispenses the sanitizer by sensing the palm of the person as soon as it comes under the nozzle automatically. The nozzle is timed by a relay to dispense required quantity of sanitizer with the flexibility of changing the discharge time and thus varying the quantity of sanitizer as per the user requirement.

Dr. Anuj Kumar Sharma and his team member Mahip Singh designed and developed the hospital assistance robot with multi-functions like:

1. Serving food and medicines in the wards.
2. Sanitize the ward using UV light and sanitizer spray.
3. Convertible to chair for moving patients from one position to other without human intervention with patient.
4. Thermal scanning of the patient. Monitoring the ward and patients with the camera installed on it



Hospital Assistance Robot: Patent filed

Rapid Multipurpose Disinfector:
Patent filedRapid Currency Sanitizing Machine:
Patent published

Dr. Anuj Kumar Sharma has filed three patents on the following products useful for prevention and protection from COVID-19 virus.

SELECTED PUBLICATIONS FROM THIS WORK

1. Vaibhav Singh, Anuj Kumar Sharma, Ranjeet Kumar Sahu, Jitendra Kumar Katiyar, Novel application of graphite-talc hybrid nanoparticle enriched cutting fluid in turning operation, Journal of Manufacturing Processes, Volume 62, 2021, Pages 378-387. IF 5.01
2. Vineet Dubey, Anuj Kumar Sharma, Prameet Vats, Danil Yurievich Pimenov, Khaled Giasin, Daniel Chuchala. Study of a multicriterion decision-making approach to the mql turning of aisi 304 steel using hybrid nanocutting fluid. [Materials](#). 2021, 14(23), 7207. IF: 3.623
3. Anuj Kumar Sharma, Arun Kumar Tiwari, Amit Rai Dixit, Rabesh Kumar Singh, "Measurement of machining forces and surface roughness in turning of AISI 304 steel using alumina-MWCNT hybrid nanoparticles enriched cutting fluid", Measurement, Volume 150, 2020, 107078. IF- 3.3.
4. Shubrajit Bhaumik, Viorel Paleu, Rajan Pathak, Rishabh Maggirwar, Jitendra Kumar Katiyar, Anuj Kumar Sharma, Tribological investigation of r-GO additived biodegradable cashew nut shells liquid as an alternative industry lubricant, Tribology International, Volume 135, 2019, Pages 500-509. IF 4.872.
5. Neha Mohanta, Rabesh Kumar Singh, Jitendra Kumar Katiyar, Anuj Kumar Sharma. (2021) A Novel Fluid-Structure Interaction (FSI) Modeling Approach to Predict the Temperature Distribution in Single Point Cutting Tool for Condition Monitoring During Turning Process. Arabian Journal for Science and Engineering. (Springer), IF- 2.334.

Research Team:



Dr. Anuj Kumar Sharma
Assistant Professor, CAS



Dr. Rabesh Kumar Singh
Assistant Professor, CAS

A Composite Sponge Structure for Oil Spill Cleanups: Saving the Environment from an Oil Spill

All over the world, the oil and gas industry produce, refines, transports, and markets trillions of barrels of crude oil and petroleum products annually as per the global demand. Oil as a commodity has dominated the world energy market for decades. As oil use roared, its refined products like gasoline, diesel, and other side products drove automobiles, ships, storage terminals, and human daily life usage. The transportation practices had been done so far without serious incidents until the oil spill disasters. The enormous exploration and production activities in the continental shelf since 1960 and the use of supertankers having more than 500,000 metric tons transportation capacity have caused oceanic oil spills- a serious environmental issue. A composite sponge is developed and investigations are done in the material synthesis laboratory, Nanotechnology program, CAS. The target of the study was:

- To maximize the recovery of spilled oil/crude oil.
- To develop an ecofriendly, effective & efficient sponge structure.
- To investigate the Trapping capability & Recyclability of developed sponge structure.

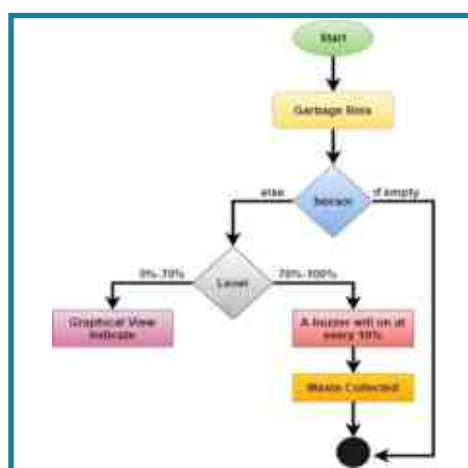
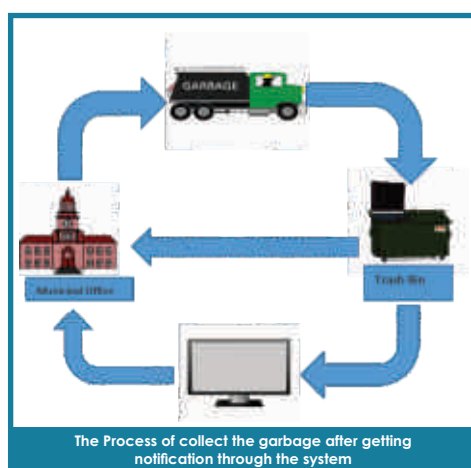
The developed composite sponges showed significant trapping capacity for diesel oil, mustered oil and crude oil. Hence, developed composite sponge can be used for effective oil spill cleanup jobs. All the developed sponges showed great recyclability in the whole cleanup job at laboratory scale. The fabricated sponge structure resulted in very good porosity

making it suitable to trap more and more oil in its pore spaces. Composite sponge showed maximum water contact angle which was act as a shuperhydrophobic and superoleophilic. There were no environmental implications observed during the whole experimental process and recovery process. Hence, the process was ecofriendly.



KusBin: A Smart Idea of Monitoring Garbage

The presence of waste in the surroundings for more than two days caused an unpleasant odor to permeate the air. This is responsible for creating an unsanitary atmosphere in metropolitan areas, and today, the polluted environment is accountable for the majority of health problems that people suffer as a result of these conditions. In order to deal with this, a framework for trash accumulation must be established. In today's environment, the Internet of Things (IoT) is used everywhere. It makes the work much more straightforward and efficient. As a result, various Internet of Things-based apps have been proposed to address the trash disposal and overflow issues. Using the Internet of Things (IoT), this study conducts a brief survey on smart garbage monitoring systems. This study also proposes a smart garbage monitoring system made up of numerous micro/nano-electromechanical devices, with the purpose of enhancing the speed and efficiency with which waste is collected and disposed of. This project is now in the works to trap the emitting dangerous gases at the precise time of their emergence into the environment using a rapid decision making technique and artificial intelligence for detection of hazardous gases in various weathering conditions.



Proposed Process of Efficient Garbage Management



Dr. Saurabh Mishra
Assistant Professor, CAS



Samarpit Verma
M.Tech EST Student



Ujjawal Singh
M.Tech EST Student

Research Publications from the Projects

1. Saurabh Mishra, Geetanjali Chauhan, Samarpit Verma, Ujjawal Singh, "The Emergence of nanotechnology in mitigating Petroleum Oil Spills" Accepted in Marine Pollution Bulletin, 2022, Elsevier Publications (IF-5.553), (SCI Indexed).
2. Samarpit Verma, Ujjawal Singh, Saurabh Mishra, "Saving the Environment from an Oil Spill: Critical Role of Digital Platforms" Presented at International Conference on Advances and Innovations in Recycling Engineering (AIR-2021), Dehradun, Accepted for Publication in Scopus Indexed Journal.
3. Ujjawal Singh, Samarpit Verma, S., Saurabh Mishra, "KusBin: A Smart Idea of Monitoring Garbage" Presented at International Conference on Advances and Innovations in Recycling Engineering (AIR-2021), Dehradun, Accepted for Publication in Scopus Indexed Journal.

Applications of Artificial Intelligence in Software Engineering

Machine Learning based Tool for Automatic Organization of Files

According to the sixth edition of DOMO's report, over 2.5 quintillion bytes of data is generated every single day, and data generation is only going to rise every single day. According to Forbes report, there are 2.5 quintillion bytes of data generated each day at our current rate, but that rate is only speeding up with the technological advancements. No doubt that technology has helped us in various ways be it to find solutions to a problem or to manage things. It provides a comfortable and convenient environment to us and makes the task easier to save time and efforts. We often download a bunch of files over the network or create a bunch of files in the system. Sometimes, it becomes difficult to find a single file from the set. A tool is developed that focuses on providing a solution for users by automatic organization of all types of files including PDF, documents, images and audio files. The tool is developed on the Django framework and in the back- end, machine learning models are used. The front end consists of HTML, CSS, Java, and BootStrap. The tool is user friendly and easy to use.

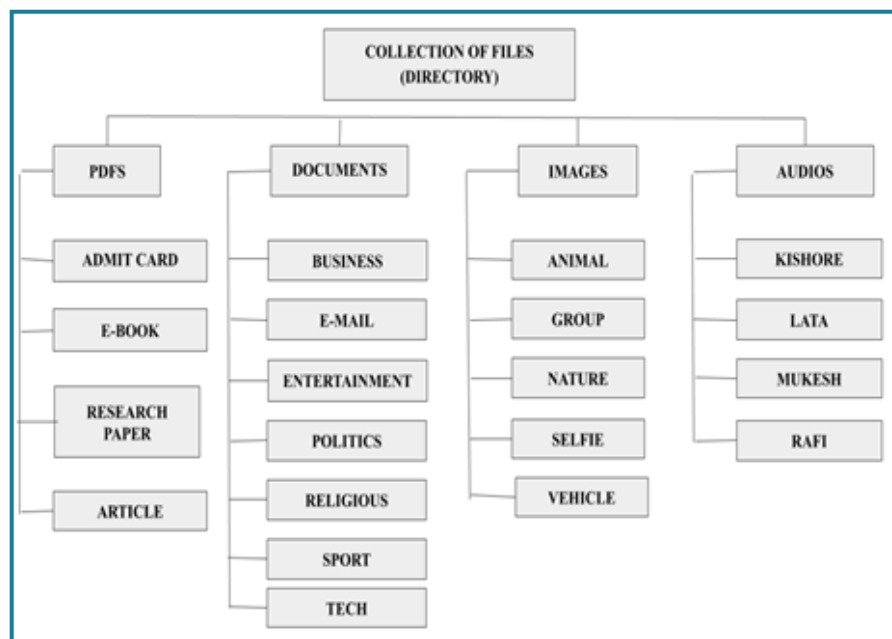


Fig. Hierarchy of Categorization of Input Directory

The tool automatically organizes all files in the input directory into a hierarchy of folders and sub-folders as shown in the figure. It then becomes easy to find a single file when a directory is organized into twenty sub-directories. The tool also provides SAVE feature to save any file in the organized directory, and SORT feature to enable sorting of the organised directory based on different criteria. This tool works magic and organizes the directory automatically for its users.

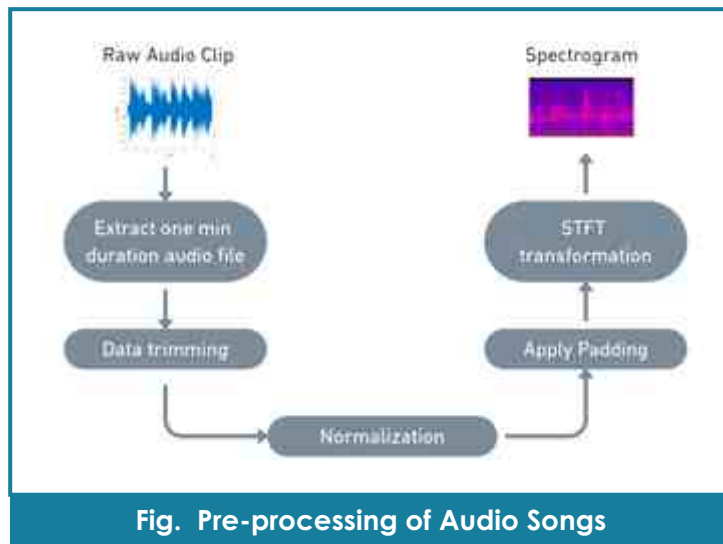
List of Publications from the Research Work:

- [1] Sengar N., Singh A., Yadav V. (2021) Classification of Documents Using Bidirectional Long Short-Term Memory Recurrent Neural Network. In: Reddy V.S., Prasad V.K., Wang J., Reddy K.T.V. (eds) Soft Computing and Signal Processing. Advances in Intelligent Systems and Computing, vol 1325. Springer, Singapore. https://doi.org/10.1007/978-981-33-6912-2_14.
- [2] Kaushik M., Rani S., Yadav V. (2021) Vocalist Identification in Audio Songs Using Convolutional Neural Network. In: Biswas A., Wennekes E., Hong TP., Wiczorkowska A. (eds) Advances in Speech and Music Technology. Advances in Intelligent Systems and Computing, vol 1320. Springer, Singapore. https://doi.org/10.1007/978-981-33-6881-1_9D.
- [3] Singh D., Bhatnagar M., Yadav V. (2022) PDF Classification Using Logistic Regression and Latent Dirichlet Allocation. In: Gunjan V.K., Zurada J.M. (eds) Proceedings of the 2nd International Conference on Recent Trends in Machine Learning, IoT, Smart Cities and Applications. Lecture Notes in Networks and Systems, vol 237. Springer, Singapore. https://doi.org/10.1007/978-981-16-6407-6_36
- [4] V. Tiwari, C. Pandey, A. Dwivedi and V. Yadav, "Image Classification Using Deep Neural Network," 2020 2nd International Conference on Advances in Computing, Communication Control and Networking (ICACCCN), 2020, pp. 730-733, doi: 10.1109/ICACCCN51052.2020.9362804.
- [5] Rani S., Kaushik M., Yadav V. (2022) Identifying Mood in Music Using Deep Learning. In: Raje R.R., Hussain F., Kannan R.J. (eds) Artificial Intelligence and Technologies. Lecture Notes in Electrical Engineering, vol 806. Springer, Singapore. https://doi.org/10.1007/978-981-16-6448-9_55

Applications of Artificial Intelligence in Music

Categorization of Audio Songs using Deep Learning

There are millions of songs in the world and numerous types of music genre exist worldwide. Manually, it is a cumbersome task to classify songs based on the vocalists. On the other hand, search algorithms can efficiently recognize a song but these algorithms use pattern matching technique in a database and therefore cannot be generalized over known songs. Music companies often face these types of challenges in automation of song classification. Hence, there is a need for artificial intelligence-enabled method to classify songs based on vocalists which can also be generalized for unknown songs. A novel Bollywood song dataset (BSdataset) is created in this work. The pre-processing of dataset involves the conversion of an audio file into a spectrogram, i.e. visual representation of frequencies of audio signal as it varies with time and then uses these spectrograms as an image to train a convolutional neural network (CNN) for classification of vocalists in an audio song.



Each genre contains several pitches of a song. It can be exhausting to extract pitches from a song manually. The work also focuses on identification of mood in a song based on genre pitch using deep learning algorithm and classifies the song on the basis of the mood like loud and relaxed. Preliminary phase involves similar conversion of a music clip into a spectrogram as shown in the figure that demonstrates the signal frequencies with time in an oriented waveform. Thereafter, it can be used as an image to convolutional neural network (CNN) for the motive of mood identification. We achieved a gratified result after testing 480 representative music clips with the CNN model and the obtained accuracy is 90%.



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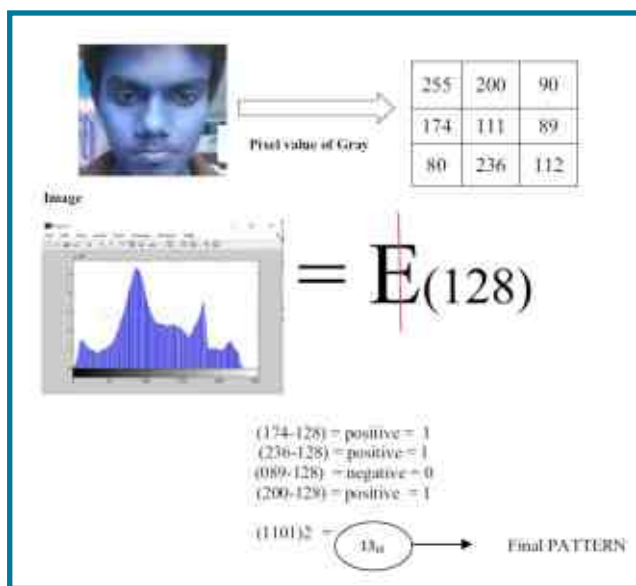
Shikha Rani
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Akriti Singh
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An improved approach towards biometric face recognition using artificial neural network

In a biometric-based security system, rather than depending on the system configuration itself, failure rate also relies upon feature extraction and its related statistics. Here, a significant approach is being presented to minimize the failure rate and maintain high recognition accuracy and uniformity for non-symmetrical feature points. This work contributes a detailed analysis of stable parameters of captured biometric feature points by using a flexible learning model named as adopted Artificial Neural Network (ANN). This work also discusses a comparative study of different global and local methods of Histogram of an Equivalent Pattern (HEP) technique for facial feature detection and extraction. The HEP is further classified by using adopted ANN model, which depends on partitioning the feature area of a predefined image. This task has been accomplished by providing the appropriate definition of local and global functions based on pixel intensities.



The literature available for face detection shows many shortcomings such as false acceptance and rejection rates. Among all defined global and local techniques, the work primarily endorsed an adopted method of Improved Local Binary Pattern (ILBP) which works on local pixel values of a facial image for feature extraction. The classification and recognition task are performed by adopted ANN for various defined global and local features. The work also derives a detailed comparison with the other existing techniques. As a result, the proposed ILBP technique ensures the consistency of acceptable results in unpredictable variations in the dataset.

Research Team :



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Ankit Kumar
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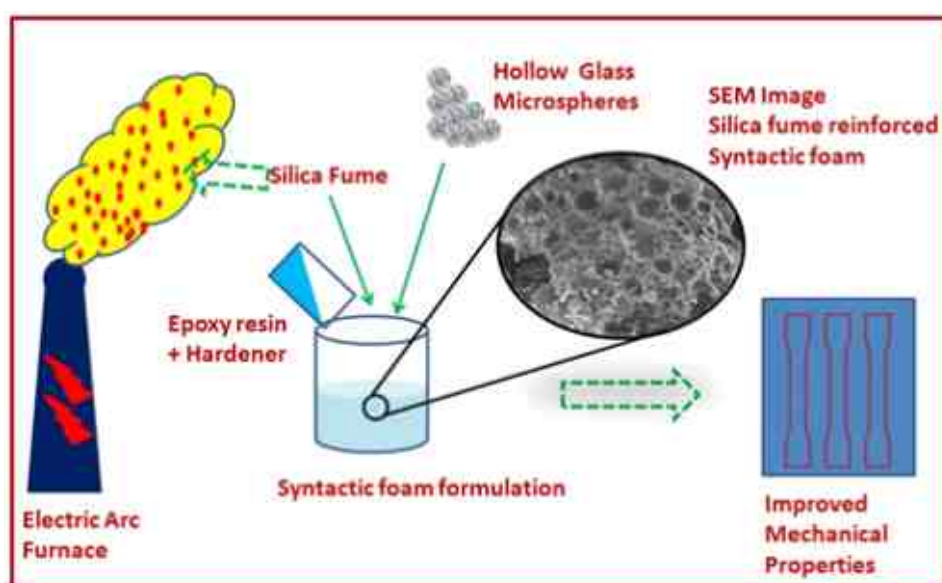
Dr. Arun Kumar
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Publication from the work:

Shubhi Srivastava, Ankit Kumar, Anupam Singh, Shiv Prakash, Arun Kumar, "An improved approach towards biometric face recognition using artificial neural network," Multimedia Tools and Applications (Accepted)(SCI indexed, impact factor- 2.757) , <https://doi.org/10.1007/s11042-021-11721-2>

Utilization of Silica fume towards reinforcing Epoxy Hollow Glass Microballoons Syntactic Foams for varied Applications

Silica fume is a by-product that is obtained during the smelting process of silicon and ferrosilicon alloys. It is regarded as a waste product but owing to the high percentage of silica (around ~99 %) this waste product has nowadays been finding wide usage in construction industry for the reinforcement of concrete to enhance the mechanical properties of concrete and preventing the early onset of corrosion in steel bars. This project was an attempt to reuse this waste product as a potential reinforcement in making epoxy hollow glass microballoons syntactic foams. Syntactic foams are used in areas where weight is a constraint. They possess high specific strength and low density which makes them suitable for defence, in particular, blast mitigation applications. They are finding increased usage as sandwiched materials for the construction of aeroplanes and submarines etc. Owing to the widespread use of syntactic foams and silica fume in the construction industry, it was



hypothesized that they be used as reinforcement (1-7 % v/v) in making epoxy hollow glass microballoons (40- 60 % v/v) syntactic foams. The specimens were prepared by stir casting approach. 40 % (v/v) hollow glass microballoons was chosen as the base composition and effect of adding silica fume (1-7 % v/v) was evaluated on this composition. Quasi-static mechanical properties of the foamed composites were determined experimentally in tension, compression and flexure mode

only and improvement in properties quantified. The change in density of syntactic foams both unreinforced and reinforced was determined by averaging the mass: volume ratio of the specimens as per ASTM D1622-98. The same is presented in Figure 1.

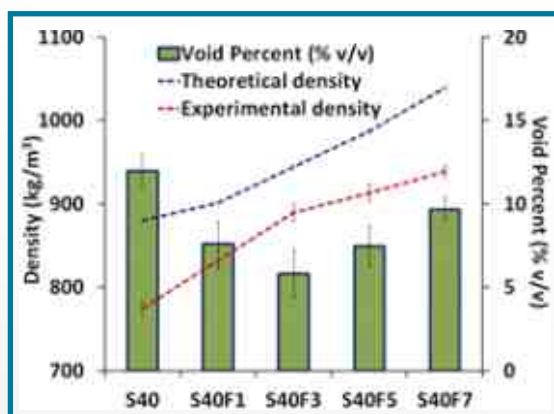


Figure 1: Theoretical and experimental densities along with associated voidage

PUBLICATION

1. Vikram Singh, V., P. Jaiswal and A. V. Ullas (2021). "Mechanical properties of silica fume reinforced epoxy glass microballoons syntactic foams." *Materials Today: Proceedings* 46: 672-676

The presence of silica fume served as load-bearing components in the matrix and promoted effective stress transfer. This resulted in substantial improvements in the mechanical properties of syntactic foam. At 5 % (v/v) loading of silica fume particles ~16 %, ~23 %, ~25 %, ~47 % and ~54 % improvements were obtained in compressive strength, compressive toughness, tensile strength, specific tensile strength and flexural strength respectively. The promising results promote the usage of silica fume as reinforcement in epoxy syntactic foams for advanced applications.

Research Team :



Veer Vikram Singh
(M.Tech Nanotech Student)



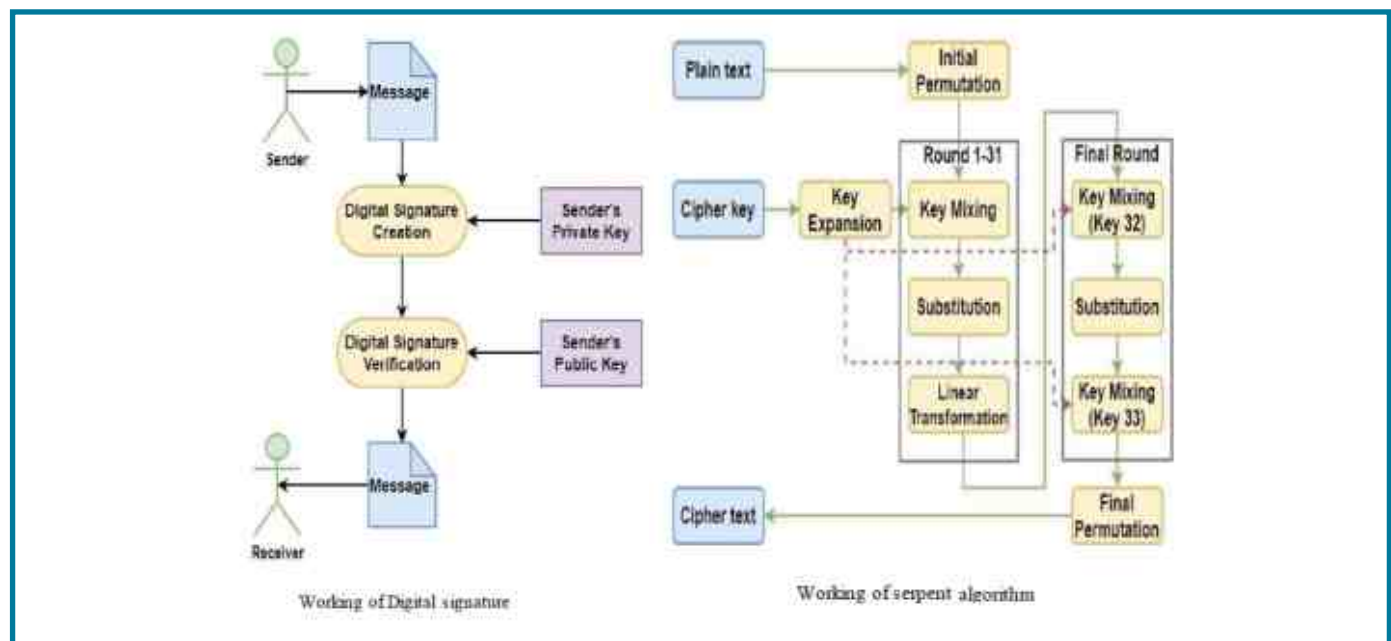
Late. Dr Piyush Jaiswal
(Assistant Professor, CAS)



Dr A.V. Ullas
(Assistant Profesor, CAS)

Handling Secure Healthcare Data Streaming using R2E Algorithm

In the present era, the transfer of information has increased using the internet, which cannot be reliable because any unauthorized user can tamper the data or obtain the unintended information, especially in healthcare. With the rise in the number of patient's health record, there's an increase in the number of data breaches. Healthcare data is something which sometimes can't be shared with unauthorized people because it stores lots of important personal and sensitive data. However, technologies such as cloud storage stored at any remote location (cloud) which is vulnerable to many cyber-attacks pose an adverse impression on the secrecy and safety of patient's healthcare records and require lots of security for protecting the medical information. So, protecting the confidentiality of the patient's health information is of utmost importance. This work tries to provide a better and more efficient solution for securing all the health care data that are stored on the cloud which protect health care data from unauthorized access from an unknown source.



Research Team :



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Rohit Kumar Pathak
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Dr. Arun Kumar
Assistant Professor, CAS

Publication from the work

Umesh Kumar, Rohit K. Pathak and Arun Kumar, "Handling Secure Healthcare Data Streaming using R2E Algorithm," 2020 International Conference on Electronics and Sustainable Communication Systems (ICESC), 2020, pp. 732-737, doi: 10.1109/ICESC48915.2020.9156006.

Signature based Authentication: A Multi-Label classification approach to detect the language and forged sample in signature

The handwritten signature is the most widely used biometric because of its low-cost and straightforward acquisition process, and also it has been used in most of the applications like finance, education, administration, etc. Signature verification is considered the most challenging field in computer vision because of its high inter-class similarity. Also, it becomes quite hard when skilled forgery has been introduced in the system. Being the cost-friendly and straightforward biometric, signature prone to forgery. Therefore, it is necessary to identify that the genuine person or the impostor attempts to access the system. Hence, it is required to build a system that will identify the forged signature from the pool of signatures. In real time scenario, we do not have forged samples at the time of training. To address the above said problems, we have proposed a multilabel system for the authentication of the signature images. In the proposed work, we have

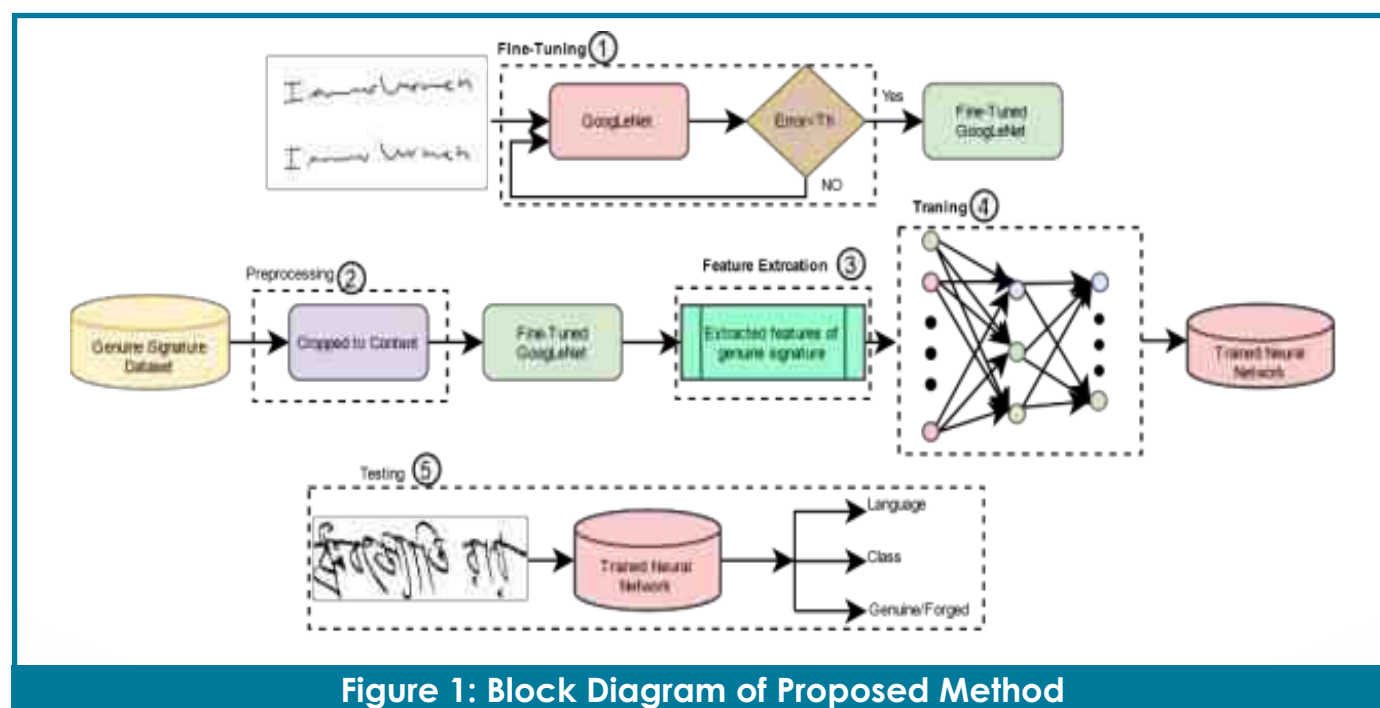


Figure 1: Block Diagram of Proposed Method

not introduced any forged signatures at the time of training, only genuine signatures are considered. Figure 1 shows the block diagram of the proposed method. In the proposed system, pre-trained GoogLeNet has been fine-tuned over the related dataset GPDS synthetic subcorpus and from BHSig260 dataset (has signatures in two regional languages i.e. Hindi and Bengali) features has been

extracted. The extracted features fed to the neural network for classification. The system generate three decisions i.e. the signature is genuine or forged, the class of the signature, and in which language the signature has been signed. The propose method is language independent and can be used in financial, legal organizations. The methods gives 98.85% accuracy over BHSig260 Dataset



Dr. Anamika Jain
Assistant Professor, CAS

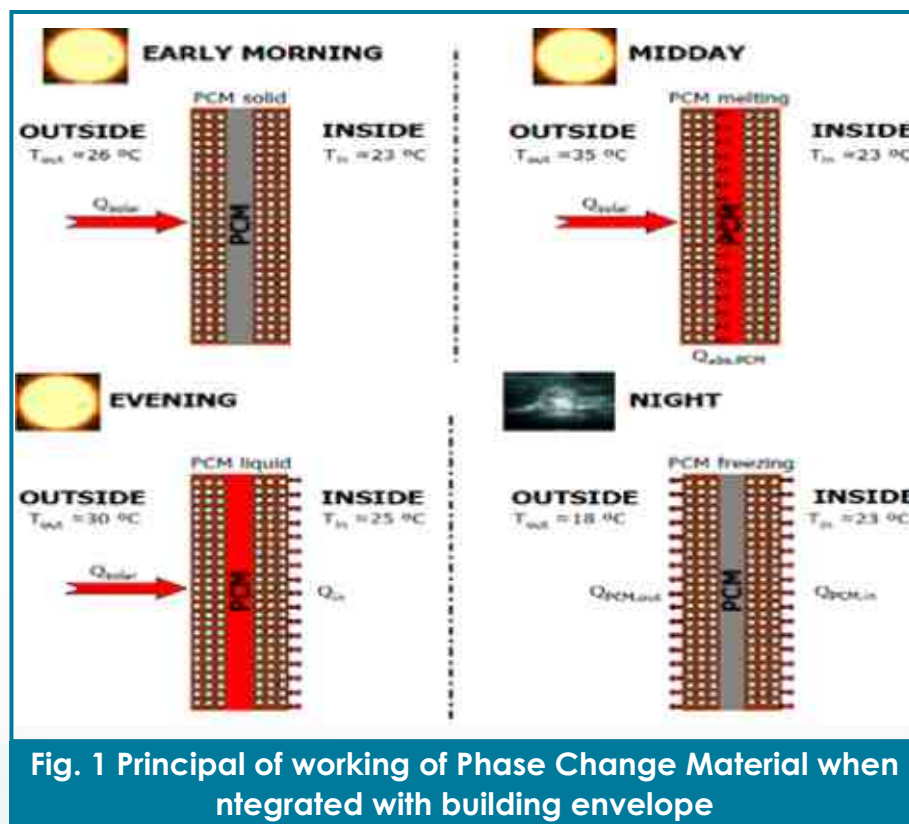
Publications from the work :

- Signature based Authentication: A Multi-Label classification approach to detect the language and forged sample in signature, Anamika Jain, Satish Kumar Singh, Krishna Pratap Singh, *Computer Vision and Image Processing, IIT Roorkee 2021.*

Thermal performance of the building envelope integrated with phase change material for thermal energy storage

One of the major sources of end-use of energy in buildings is consumed in space cooling for improving indoor thermal comfort. Climate extremities like heat waves and extreme temperature events have further pushed the demand for energy in buildings. Enhancing the thermal energy storage capacity of the building envelope by incorporating PCM is one of the sustainable methods of improving the indoor temperature profile as shown in fig. 1.

1. PCM can store multiple times more energy than conventional building materials like concrete, bricks, tiles, and cement plaster.
2. PCM are available over a wide range of temperatures and thus increase their applicability to various types of climatic conditions/seasons.
3. Three types of PCM incorporation methods in the building element are identified i.e. Microencapsulation, Macroencapsulation, and Shape Stabilized PCM. Out of these three Microencapsulation and Macroencapsulation techniques are widely used due to minimizing leakage and ease in incorporation in the building envelope.
4. PCM when embedded in the building material improves its latent heat storage capacity. Increasing the quantity of embedded PCM will



Dr. Pushendra Kumar Singh Rathore
Assistant Professor, CAS

Research Publication from above research

Rathore, P. K. S., Gupta, N. K., Yadav, D., Shukla, S. K., & Kaul, S. (2022). Thermal performance of the building envelope integrated with phase change material for thermal energy storage: an updated review. *Sustainable Cities and Society*, 79, 103690. **Impact factor: 7.58.**

Analysis of solar still using Pyrex glass quantum dot in tropical climate

Clean freshwater is one of the essential ingredient for human life, but 1.1 billion people lack access to water and 2.7 billion experience water scarcity. A new World Bank reports finds that water scarcity, exacerbated by climate change, could hinder economic growth, spur migration, and spark conflict. Therefore, to mitigate this challenge Centre for Advanced Studies, Lucknow has come forward and conducted a research on sustainable way of producing fresh potable water for drinking as shown in fig. 1.

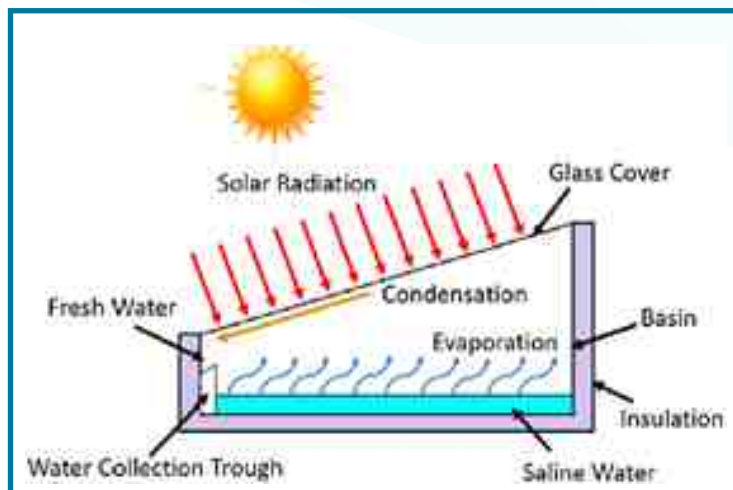


Fig.1 Schematic of Solar Still

This method of converting saline ocean water into pure drinking water is eco-friendly, clean, and cheap. The experimental set up consist of Solar Still, Pyrex Glass Quantum Dot, and thermocouples. Experiments were conducted for a depth of 0.02 m for solar still with quantum dot and without quantum dot. The results revealed that for 0.02 m of water depth and 1 m² of basin area, the percentage increase in distillate output using Pyrex glass powder quantum dot is 29.36% experimentally and 29.94% theoretically. Furthermore, the net percentage increase in solar still efficiency by using Pyrex glass quantum dot is 74.74% experimentally and 61.03% theoretically.



(a)

(b)

**Fig. 2 Experimental set up of solar still
(a) without Quantum Dot (b) with glass powder Quantum dot**



Dr. Pushpendra Kumar Singh Rathore
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Research Publication from above research

Singh, PK, Rathore, PKS, Shukla, SK. Experimental and numerical analysis of solar still using Pyrex glass quantum dot in tropical climate. *Int J Energy Res.* 2021; 1- 15. **Impact factor: 5.1.** doi:10.1002/er.7214

Nanomaterials for Photonic and Energy Storage Applications

Nanomaterials are the class of materials in which at least one dimension is lesser than 100 nm and display distinct properties as compared to that of their bulk counterpart. Their properties are strongly dependent on their particle size, shape, crystal phase, and composition. Because of their fascinating properties, they find applications in various domains of science, engineering, environment, and medical sciences such as solar energy harvesting, energy storage devices, drug delivery, bioimaging, photo-thermal therapy, and catalysis industry. Currently, at Nanotechnology department, Centre for Advanced Studies (CAS), we are focused on synthesizing high quality nanoparticles with desired morphology, phase, and composition for energy storage applications such as supercapacitor and batteries. In this line, we are developing inorganic nanoparticles based nano-catalyst such as Fe_2O_3 , Co_3O_4 , MnO_2 , NiO to characterized them using X-ray diffraction (XRD), scanning electron microscope (SEM), energy dispersive (EDXA), and cyclic voltammetry for structural, compositional, and electrochemical properties.



Dr. Chandresh Kumar Rastogi
Assistant Professor CAS

Publication from above work

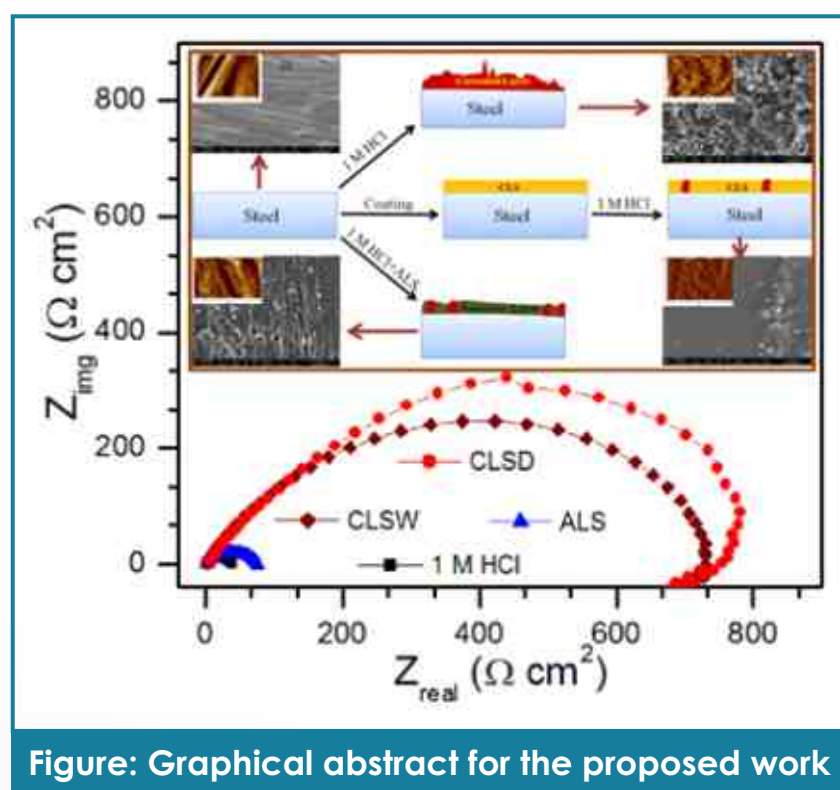
C. K. Rastogi, "Synthesis Techniques for Controlled Preparation of Up-Converting Lanthanide-Ions Doped Fluoride Nanophosphors" Chapter in a book titled "Synthesis strategies for advanced materials", (2021) Springer Publishing Company, ISBN 978-981-16-1892-5

"Nanotechnology is the idea that we can create devices and machines all the way down to the nanometer scale, which is a billionth of a meter, about half the width of a human DNA molecule"

Paul McEuen

Lemon seeds as green coating material for mitigation of mild steel corrosion in acid media

Natural materials are the subject of high priority for corrosion scientists, now a day. Various natural materials have been used to date for corrosion protection of metals. However, their utilizations have been done either in inhibition (mostly) or coating applications (only in composite form). In present work, lemon seed (alone) is used for inhibition as well as coating protection of mild steel (MS) corrosion in acid media. Aqueous extract of lemon seed (ALS) is used as inhibitor, whereas chloroform extract of lemon seed (CLS) is used to form wet (CLSW) and dry (CLSD) coating on MS substrate. Characterizations of ALS, CLSW and CLSD have been done by UV-visible spectroscopy, FTIR spectroscopy, HRSEM, EDAX and AFM analysis. Corrosion testing of both extracts has been performed in 1 M HCl by ac impedance measurements (AIM) and Tafel polarization curves (TPC). The results show that ALS gives very low inhibition of MS corrosion ($\sim 45\%$), while MS coated with CLSD/CLSW provide high efficiency protection ($\sim 98\%$) to the substrate.



Publication

Shweta Pal, **Gopal Ji**, Hassane Lgaz, Ill-Min Chung, Rajiv Prakash, Journal of Molecular Liquids, Volume 316, 2020, 113797, ISSN 0167-7322, DOI: 10.1016/j.molliq.2020.113797., IF-6.165, Q1, SCI Journal



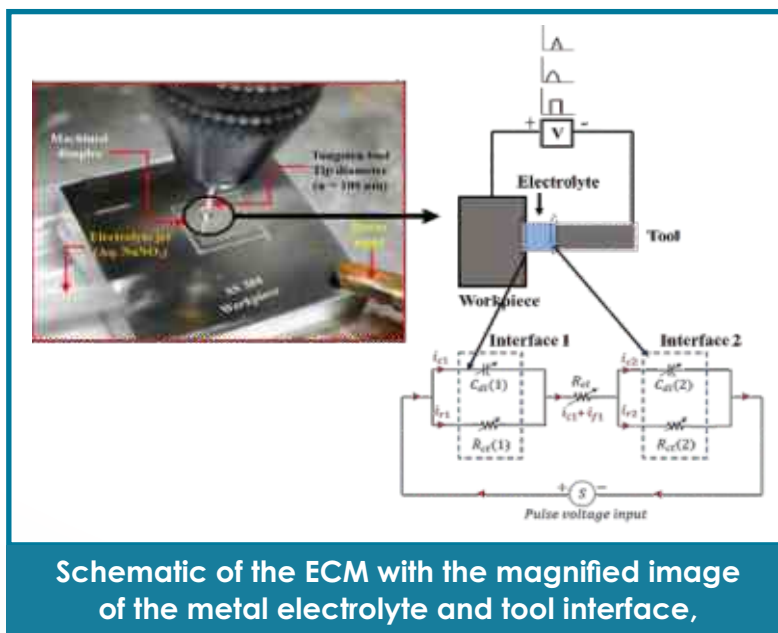
Dr Gopal Ji
(Assistant Professor, CAS)

**"I think fragility drives innovation, just like other constraints do.
One of the only ways to get out of a tight box is to invent your way out"**

Jeff Bezos

Effect of Voltage signal waveform for electrochemical machining.

The electrochemical machining (ECM) is a process of removing the metal with the help of potential bias shown in figure. The amount and the shape of the metal piece remove depends on the waveform of the potential. The selection of voltage waveform is critical to achieving the desired machining efficiency and accuracy in ECM. The electrochemical impedance spectroscopy (EIS) estimates the different physicochemical processes occurring at the metal electrolyte and tool electrolyte interface. These physicochemical processes can be correlated in the electrical circuit elements like resistance and capacitance occurring at the two interfaces shown in Figure. The EIS is conducted at the different potentials of 1 to 5 V with a 1V difference. The electrical elements, resistances, and capacitances are fitted to the function of the voltage. The electrical circuit with the circuit element as a function of voltage will estimate the theoretical current obtained from the input voltage waveform. The input voltage is in the form of the half-wave rectangular (RVW), sinusoidal (SVW), and triangular (TVW) waveform. The variation of current with time is



showe that for the RVW, the ECM gives maximum current, showing maximum efficiency with the maximum material removed. It is due to the voltage staying longer at the maximum value. On the other hand, the TVW gives the lowest current showing the maximum machining accuracy. Thus, for a good ECM accuracy, we should go for TVW, while for best ECM efficiency, one must prefer RVW.



Dr. Gyanprakash D. Maurya
Assistant Professor, CAS

Research Publication from the above work

Sharma, V., Maurya Gyanprakash D., Gupta, P. and Ramkumar, J., Analysis of circuit current in electrochemical micromachining process under the application of different waveforms of pulsed voltage. Journal of Manufacturing Processes, 2022, 75, pp.110-124.

**“To succeed in your mission, you must have single
- minded devotion to your goal”**

— Dr. A.P.J. Abdul Kalam



Analysis of hot dip galvanised steel wires exposed in different environmental sites

This work is focused on investigation of hot dip galvanized steel (HDGS) wires exposed in Belgium and Indonesia for 4.5 years and correlation of corrosion products formed on the exposed wires with their corrosion properties was studied. Corrosion products were identified with local Raman analysis and correlated with corrosion activity based on odd random phase electrochemical impedance spectroscopy (ORP-EIS) and Tafel behaviour. Raman spectroscopy results confirmed that corrosion products formed in both atmospheric conditions differed either in quantity and/or in chemical nature, which indicated that “aged in Belgium” and “aged in Indonesia” wires did not have similar surface compositions. The comparison of the instantaneous corrosion behaviour of pristine and aged wires based on corrosion currents (polarization curve) and impedance values in 0.1 M NaCl and 0.1 M Na₂SO₄ declared that aged in Belgium (AIB) and aged in Indonesia (AII) wires exhibited different corrosion behaviour, which confirmed that corrosion products formed in the environments of the exposed wires were dominating their corrosion behaviour

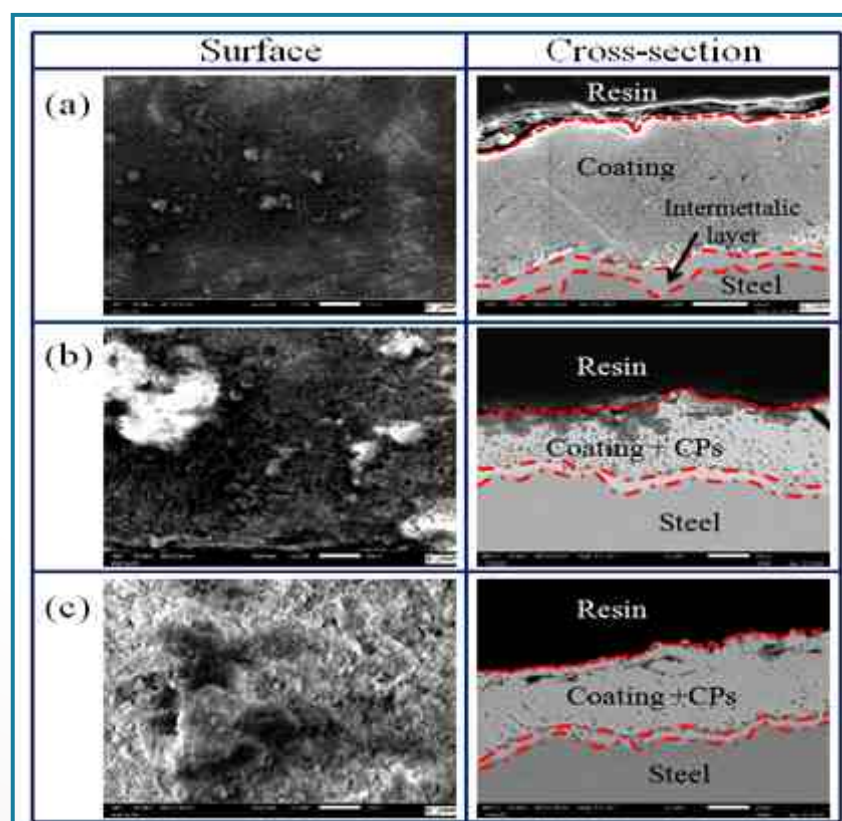


Figure: Surface and cross sectional examination of exposed steel wires.

Publication

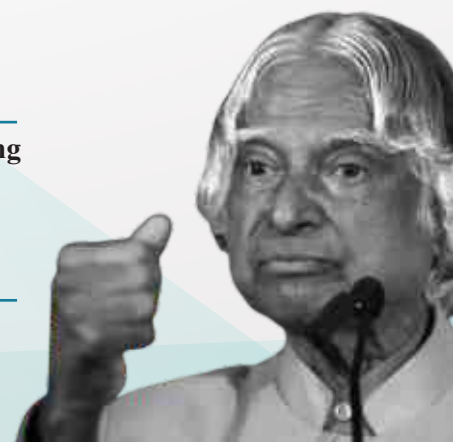
Gopal Ji, Kitty Baert, Bart Allaert, Annick Hubin, Herman Terryn, , **Corrosion Engineering, Science and Technology** 55 (7), 562-573, IF-2.087, Q2, SCI Journal



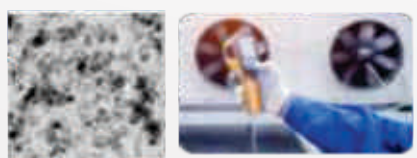
Dr Gopal Ji
(Assistant Professor, CAS)

“Creativity is seeing the same thing
but thinking differently.”

— Dr. A.P.J. Abdul Kalam



Major Funded Research Projects

Sr. No.	Title of Project	Funding Agency	Name of PI/ Co-PI	Amount / Duration
1	Intelligent Stethoscope: A low cost device based on body auscultation for Early Medical diagnosis of Heart, Lung and Prenatal health 	TDT Division, Biomedical Device and Technology Development (BDTD), Department of Science and Technology, Gov. of India	Prof. M.K. Dutta (PI)	34.45 Lakh 2 years
2	Assistive device to impart perceptual ability to visually-impaired using intelligent scene captioning 	SEED Division, Technology Interventions for Disabled and Elderly Programme, Department of Science and Technology, Gov. of India	Prof. M.K. Dutta (PI)	39.25 Lakh 3 years
3	Design and Development of Artificial Intelligence based Screening Tool for Automatic Diagnosis of Osteoporosis in Women 	Biomedical Device and Technology Development (BDTD), Department of Science and Technology, DST Gov. of India.	Prof. M.K. Dutta (PI) Collaborator: KGMU, Lucknow.	74 Lakhs 3 years (Sanctioned, Fund release under Process)
4	Development of sustainable hybrid Nano-lubricant with improved Tribological properties for green machining 	SERB (statutory body of DST Govt. of India	Dr. Anuj Kumar Sharma (PI) Co-PI Prof. J. Ramkumar IIT Kanpur	18.30 Lakh 3 years

PATENTS



Sr. No.	Title of Patent	Prototype	Name of Inventors
1	"A System and Method for Machine Learning Based Automatic Detection of Abnormality from Cardiac Sound" Application number: 201911033584 Dated: 21.08.2019, Status: Applied and under examination.		M.K.Dutta, Anjali Yadav, Abhishek Kaushal
2	Multi-Mode Real-time Object Detector and Obstacle Aware Assistive System for Visually Impaired Person (VIP), Application number: 201911047766, Dated: 22.11.2019, Status: Applied and under examination.		M.K. Dutta, Rakesh Chandra Joshi, Samiya Yadav
3	Artificial Intelligence based System and Method for Multi-Class Classification of Heart Diseases, Allotted Application number 201911047767 Dated: 22.11.2019, Status: Applied and under examination.		M.K. Dutta, Shivansh Dubey, Raghuvendra Pratap Tripathi, Anjali Yadav
4	Automatic Detection of Lungs Disease using Machine Learning from Respiratory Sound, Allotted Application number: 202011009043, Dated: 03.03.2020, Status: Applied and under examination		Neeraj Baghel, M.K. Dutta
5	Artificial intelligence based COVID-19 screening tool based on analysis of X-ray images, Application no. 202011024438 dated: 10.06.2020, Status: Applied and under examination.		Rakesh Chandra Joshi Samiya Yadav M.K.Dutta, Vinay Kumar Patilak, M.L.Bhatt Anit Parihar Hardeep Singh Malhotra
6	"System and method for sanitizing currency note(s)" Application number: 202011035026 Dated: 14.08.2020, Status: Published with no. E-12/948/2020/DEL and under examination.		Anuj Kumar Sharma, Mahip Singh
7	"Application number: 202011037517 Dated: 31.08.2020, Status: Applied and under examination		Anuj Kumar Sharma, Mahip Singh
8	"Oxygen monitoring & control, Multi-Functional convertible Robot for hospitals". Filed and waiting for application number		Mahip Singh, Anuj Kumar Sharma, Amit Rai Dixit
9	"TIPI- steel and mining waste management: intelligent technology and process management for steel industry and mining waste". Application number: 202011047142 Dated: 29.10.2020, Status: Applied and under examination		Navneet Kumar, Anuj Kumar Sharma, Brijesh Singh, Yatender Chaturvedi , Sanjeev Kumar Sharma, Rakesh Kumar Yadav

Research Seminars

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Seminar -1

Deep Learning Based Framework for Microscopic Cell Images Classification using Multi-Level Ensemble

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Ritesh Maurya
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Seminar -2

Deep Machine Learning in Time Series: Cardiac and Pulmonary Signals analysis using CNN models

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SPEAKER
Neeraj Baghel
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Seminar -3

Deep Learning in Medical Imaging: Automated Detection of COVID-19 from Chest X-Rays using deep convolutional neural network

DATE & TIME:
22nd January 2021,
4:00 PM.

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SPEAKER
Saumya Yadav
Junior Research Fellow, Centre for Advanced Studies (Till Dec, 2020)

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Research Seminar Series
Seminar - 4

Deep Learning in Computer Vision with Sensors: Application in Assistance for Visually Impaired People

DATE & TIME:
27th January 2021,
4:00 PM.

PLATFORM & WEB-LINK
Google Meet :
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SPEAKER
Rakesh Chandra Joshi
Research Scholar, Centre for Advanced Studies AKTU, Lucknow

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Research Seminar Series
Seminar -5

Cost Function Optimization Strategies in Deep Neural Network

DATE & TIME:
3rd February 2021,
4:00 PM.

PLATFORM & WEB-LINK
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SPEAKER
Kumud Patel
Research Scholar, Centre for Advanced Studies AKTU, Lucknow

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Research Seminar Series
Seminar -6

Network Optimization for Deep Learning Models

DATE & TIME:
10th February 2021,
4:00 PM.

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SPEAKER
Chhavi Bajpai
Research Scholar, Centre for Advanced Studies AKTU, Lucknow

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Seminar -7
Brain-Computer Interaction using Artificial Intelligence

DATE & TIME:
17th February 2021,
4:00 PM.

PLATFORM & WEB-LINK
Google Meet :
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SPEAKER
Juwairiya Siraj Khan
Junior Research Fellow,
Centre for Advanced Studies,
AKTU, Lucknow



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Research Seminar Series

Seminar -8
Robotic fish: Design and swimming mechanism

DATE & TIME:
16th April 2021, 4:00
PM.

PLATFORM & WEB-LINK
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meet.google.com/zut-phpe-tmf



SPEAKER
Mithilesh Kumar Koiri
(Research Scholar,
Centre for Advanced Studies, AKTU
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Research Seminar Series

Seminar -9
Deep Generative Models in Deep Learning

DATE & TIME:
30th November 2021,
4:00 PM.

PLATFORM & WEB-LINK
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SPEAKER
Chetana Singh
Research Scholar
Centre for Advanced Studies,
AKTU, Lucknow



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Research Seminar Series

Seminar -10
Ensemble Learning based Classification in Artificial Intelligence and its Applications

DATE & TIME:
3rd December 2021,
4:00 PM.

PLATFORM & WEB-LINK
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SPEAKER
Karnika Dwivedi
Research Scholar
Centre for Advanced Studies
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Research Seminar Series

Seminar -11
Explainable AI in Healthcare

DATE & TIME:
18th January 2022,
4:00 PM.

PLATFORM & WEB-LINK
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SPEAKER
Vinay Priy Mishra
Research Scholar
Centre for Advanced Studies
AKTU, Lucknow



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Seminar -12
Corrosion Prevention of Steel using Arbi Leaves

DATE & TIME:
22th January 2022,
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SPEAKER
Vinit Kumar Jha
M.Tech (Final Year)
Centre for Advanced Studies
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Selected Publications

1. Chandrabhan Patel, Biswajit Mandal, Rohit G. Jadhav, Tapas Ghosh, Mayank Dubey, Apurba K. Das, Myo Than Htay, Victor V. Atuchin, Shaibal Mukherjee, "S, N Co-Doped Carbon Dot-Functionalized WO₃ Nanostructures for NO₂ and H₂S Detection", Accepted in ACS Applied Nano Materials, January 2022, Impact Factor: 5.09.
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